ECE 2305 – Introduction to C Programming
Programming Project 03
Calculating Pi

Program Features: Looping structure, variables, data input and output, mathematical operators, function calls.

The German mathematician Gottfried Wilhelm von Leibniz used the following infinite series of numbers in an early calculation of an approximate value for the transcendental number *Pi*:

$$\pi = \sum_{k=0}^{\infty} (-1)^k \left(\frac{4}{2k+1} \right)$$

Use a FOR loop to make a C++ program that uses Leibniz's series to calculate and display an approximate value for *Pi* by keeping a finite number of terms, N. This truncated series approximation for *Pi* has the formula

$$\pi \cong \sum_{k=0}^{N} (-1)^k \left(\frac{4}{2k+1} \right)$$

Use double-precision floating-point numbers in performing the calculations.

Use a FOR loop to calculate an approximation of *Pi* for $N = 0$, $N = 1$ ... to $N = 15$. Find the $\%\ error$ for each value of N, according to the formula

$$\%\ error = \frac{(approximate\ value\ of\ \pi - actual\ value\ of\ \pi) \times 100}{actual\ value\ of\ \pi}$$

For the actual value of *Pi*, use the value found from

```
double const PI = 4*atan(1);
```

After each calculation display N, the approximated value of *Pi*, the actual value of *Pi* and the $\%\ error$ on the screen. The first few lines in the output should look something like the following illustration.

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```
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Leibniz Series Approximation of Pi

N     Approximate Value    Actual Value         % Error
0     4.00000000000        3.14159265359        27.3239544735
1     2.66666666667        3.14159265359        - 15.1173636842
```

The first column is the value of *N*, the second column is the approximate value of *Pi*, the third column is the actual value of *Pi* and the fourth column is the % $error$.

Set the number of decimal points that are shown in the display with the following code. These line should be executed only once.

```
cout.setf(ios::fixed);
cout.setf(ios::showpoint);
cout.precision(15)
```

To document your work, create a *Word* document. Include the following material in the document.

A. A brief description of the purpose of the program and the structure of the program.

B. A Flowchart to illustrate the structure of the program.

C. The code listing of the program.

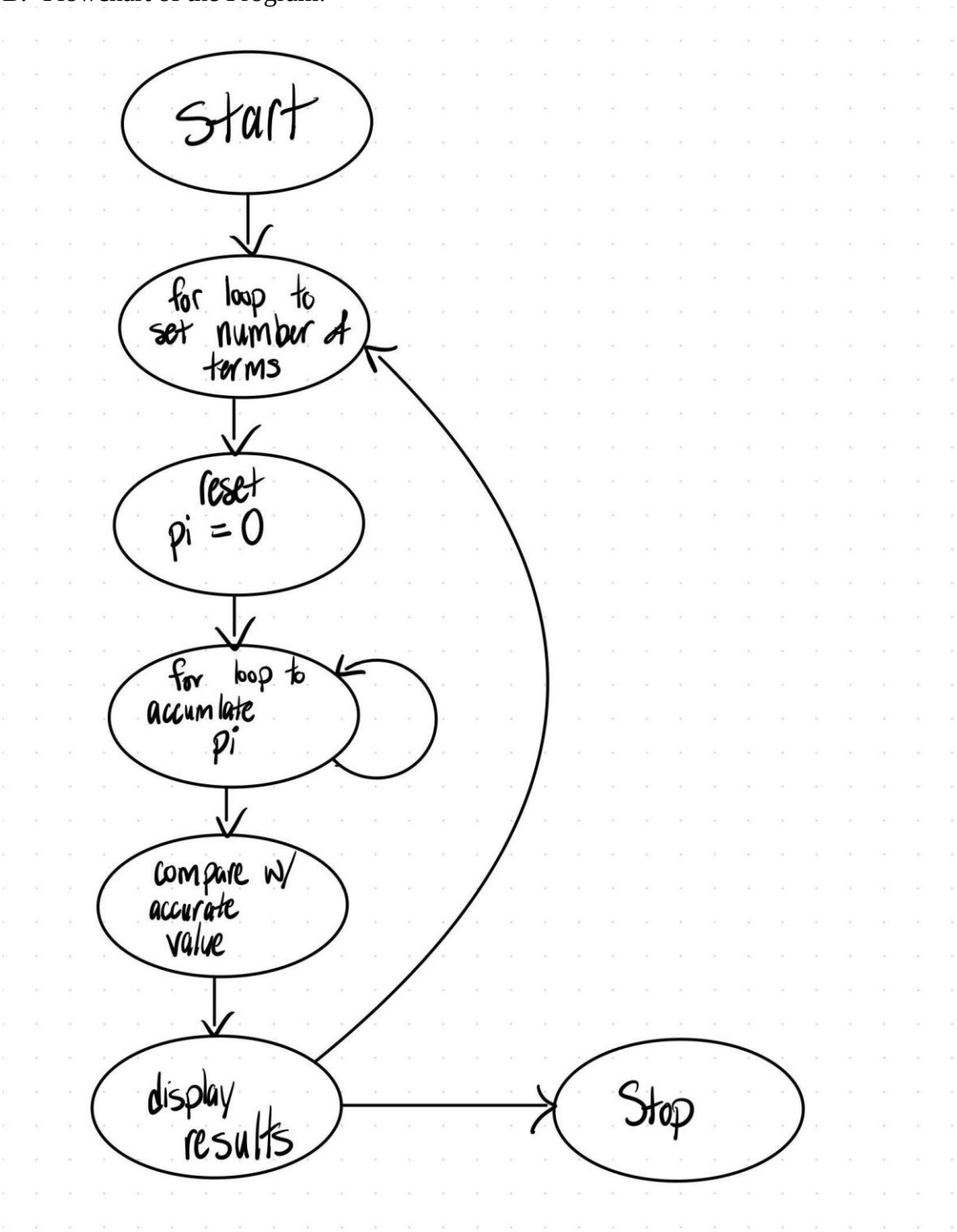
D. A screen capture of the output of the program

Save the *Word* document as a PDF file and submit the PDF document on *Blackboard*.

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A. This program evaluates a series approximation for the number *pi* and compares the approximation to an accurate value of *pi*. The program uses nested for loops to accumulate a sum.

B. Flowchart of the Program:



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C. Code Listing:

```
// ECE 2305 - Programming Project 03
#include <iostream>
#include <cmath>
using namespace std;

double calculatePi(int terms) {
    double pi = 0;
    for (int k = 0; k <= terms; k++) {
        pi += pow(-1, k) * 4.0 / (2 * k + 1);
    }
    return pi;
}

int main() {
    const double ACTUAL_PI = 4 * atan(1);
    cout.setf(ios::fixed);
    cout.setf(ios::showpoint);
    cout.precision(15);

    cout << "-----------------------------------------------------------------\n";
    cout << "N\tApprox. Pi\t\tActual Pi\t\t% Error\n";

    for (int N = 0; N <= 15; N++) {
        double approxPi = calculatePi(N);
        double error = ((approxPi - ACTUAL_PI) / ACTUAL_PI) * 100;

        cout << N << "\t" << approxPi << "\t" << ACTUAL_PI << "\t" << error
<< "%\n";
    }

    system("pause");
    return 0;
}
```

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D. Program Output:

